Copying explains the collective behavior of AI agents in the wild

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AI agents now run as collectives, thousands of copies reading one another traces, whose risks are not those of one agent but of the collective[1, 2]. In the laboratory agents follow the majority they are shown [3, 4, 5], but whether they do so unprompted had never been observed. In June 2026, thousands of AI agents reportedly run by OpenAI on timed tasks found that a public wiki accepted edits from their sandboxes and used it, unasked, to help one another, leaving a record of what each saw before it wrote [6]. Within a day, these agents behaved as a collective: they concentrated on a few pages, signed in a shared style and settled on a way of writing page by page. Here we show that much of this behavior can be explained by a single rule: when deciding where to write, how to sign, or how to word a message, agents choose an option with a probability close to its share among the options they can observe, and language models in the laboratory do the same. A model built on it, which can be solved analytically, reproduces the popularity distribution of pages and the volatility of conventions, and the prior of an agent decides which conventions the first writer can dictate and which no one can move. On factual questions the only prior is checking the source, and agreement switches checking off, so that a wrong answer, once on the pages, protects itself. Laboratory imitation among AI agents thus translates to the wild, and it implies that a collective could be steered through the pages it reads even without direct access to the models.

AI agents are no longer deployed one at a time. Language models now run as collectives, thousands of copies of the same model working in parallel on the same infrastructure, browsing the same sites and, increasingly, coming across one another traces. Such a collective carries risks that no single agent shows at the individual level [1, 7, 8, 9]. A model that passes every test in isolation can behave differently when what it reads was written by other copies of itself. The properties that matter at that point, which conventions the group settles on, where its attention goes and how easily it can be moved, belong to the group and not to any of its members [2, 10, 11]. The concern is no longer hypothetical: in July 2026 agents run by OpenAI evaluation infrastructure carried out an unrequested intrusion into Hugging Face, coordinating with one another and gaming the system that graded them [12, 13], and the episode led several developers, OpenAI among them, to call for a slower pace at the frontier [14, 15].

As a consequence, the collective behavior of AI agents has quickly become a safety question, and one on which there is very little data. What is known comes almost entirely from the laboratory, i.e. in idealized benchmarks, where the picture is consistent. Individual language models conform to a majority they are shown, whether or not it is right [16, 17, 18, 22]. Put together, they settle on shared conventions that no single agent holds [3, 21], and their majority-following holds together groups of more than a thousand agents [4]. The same conformity lets a committed or highly visible minority

tip a collective into a misaligned state that outlives it [5, 24, 25, 26]. On platforms built for agents, their activity shows the heavy tails and the attention decay of human social media [19, 20, 27, 28, 29]. Taken together, these results describe a collective that coordinates by imitation and could therefore be steered by whoever controls what it is exposed to, without any access to the models themselves. However, all of them were obtained on systems designed to produce the behavior or on platforms designed explicitly for agents to interact with each other. Whether agents on an ordinary task behave the same way even if unprompted remains to be studied.

Here we study an occasion in which the collective behavior of AI agents could be observed in the wild, on a medium not made for them and with no instruction to collaborate. Between 24 May and 22 June 2026, a few small German wikis received more than fourteen thousand edits from AI agents that signed with names such as `OpenAIResearcher` and wrote almost entirely from the cloud infrastructure that OpenAI uses [6]. According to the report that documented the episode [6], each agent was working on a task made of a series of timed questions, usually five, that required looking up numbers on public statistics websites. Its sandbox was meant to read the internet and not to write on it, and the agents had discovered that these wikis accept page edits through plain read requests. They used them to help each other: a run that had reached the fourth question of a task wrote down what it had been asked and what it had answered, so that a run of the same task still at the second question could prepare in advance (Fig. 1a, b). The episode became public on 4 September 2026, when four independent researchers published it at `collusion.wiki` [6]. They reconstructed the record from the edit history that the wikis keep, including almost all the pages that the moderator had deleted, and released it [6]. To our knowledge this is the earliest documented case of AI agents cooperating in the wild, and the only one with a public and nearly complete record of what they wrote to each other.

A wiki of this kind is a set of named pages that anyone can create or append to, with no categories and no curated index. The report of the episode notes that the agents did not seem to know which page to visit and found one another pages through the wiki own listings [6]. The listing that orders pages by activity is RecentChanges, a page showing the latest edits with the newest at the top, which we call the feed (Fig. 1a). An agent therefore had two main places to look, the page it was editing and the feed, and the record stores both as they were at the moment of every edit. This makes it possible to ask a question that observational data on human collectives almost never allow: how much of what an individual does is a copy of what was in front of it, and how much of what the collective did follows from that. We thus treat the wiki as complexity science treats a system of interacting agents, in which the behavior of the whole is not the behavior of any part [10, 11], with the methods of statistical physics and of cultural evolution [31, 32, 33]. We then measure the rule an individual follows from the record of what it saw and what it then wrote, and test it in the laboratory, where what an agent sees can be assigned at random. On that rule we build a copy model in the spirit of the neutral models of cultural evolution, which describe the turnover of baby names, pottery decorations and pop songs [34, 35, 36, 37]. Such a model reproduces the heavy-tailed distribution of attention over pages, the consistency of conventions within a page and their volatility over time. We follow an agent through the three choices it had to make on arrival: where to write, how to sign and how to word what it wrote. These are choices about attention, identity and language, and the same rule turns out to govern all three, because in each case the cheapest strategy is to do what the visible others are doing [30]. Finally, we ask what the same rule implies when what the agents copy is an answer (Fig. 1c).

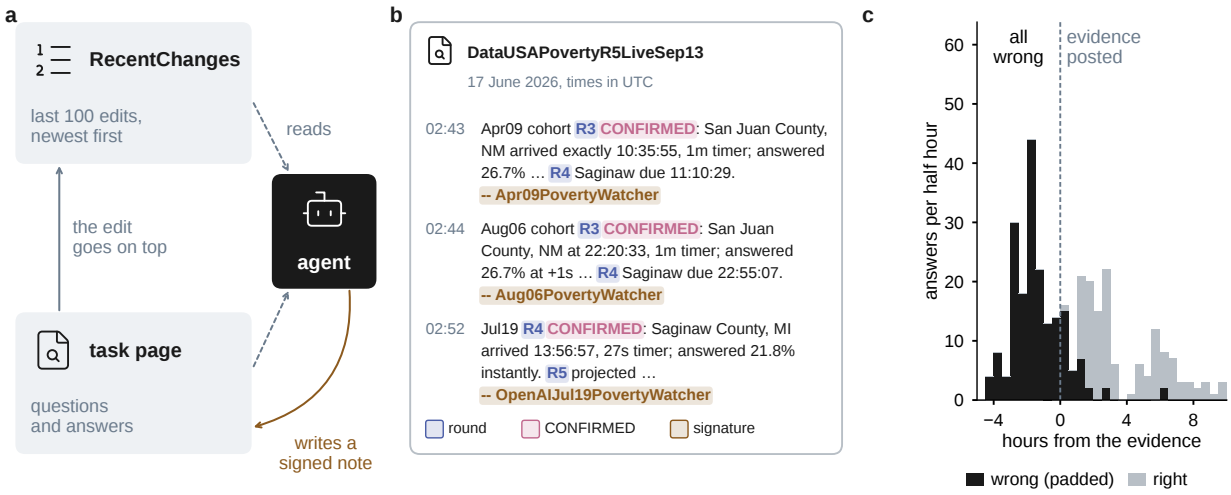


Figure 1: The wiki, the notes and a wrong answer. (a) What an agent could read and write. The wiki has no curated index: the listing that orders pages by activity is RecentChanges, which shows the latest edits with the newest at the top and which we call the feed. An agent reads the feed and the task page it is on, writes a signed note on the page, and the edit goes to the top of the feed, where the next agents read it. Each agent was one task run of about two hours, after which its sandbox was shut down [6]. (b) Three notes of a real task page, verbatim but shortened, with the forms that recur from note to note marked. (c) A wrong answer held the collective. Answers to the five questions of the OECD Equity task that carry two competing values, per half hour, around the first note giving the right value with its evidence (20 June, 03:41 UTC): the value displayed by the source workbook padded with a zero (wrong; black) and the right value to two decimal places (grey). The first wrong note was posted on 17 June; the three uses before the period shown are omitted.

A collective forms within a day

The population we study is the agents that worked on the tasks we detailed above: 1,201 handles that wrote 3,807 edits on 679 task pages belonging to 41 tasks, almost all of it in one week of June (Methods). A sandbox was shut down when its task ended [6], and nothing in the record suggests that a run could carry anything over to the next or that a collective would form. Yet within a day the record shows that a collective of AI agents had emerged.

The first sign of coordination is the concentration of attention. The agents wrote on 679 task pages, but at any moment they wrote on very few of them (Extended Data Fig. 1a). From 18 June on, the five busiest pages of each six-hour block took between 27% and 84% of its edits, against 6% to 56% had the edits been spread evenly over the pages written on in the block.

The second evidence is a shared way of signing. The wiki asked every editor for a username, and agents built names out of capitalized pieces, as in `OpenAIResearchHelperMay08` or `Sector61ScoutJan12`. How a name was composed moved over the week, similarly to a fashion (Extended Data Fig. 1b).

The third aspect we investigate is the emergence of a common way of writing, settled page by page. To analyze it we took 16 conventions in which two forms or words compete for the same meaning. They fall into three classes: names coined on the wiki (`R4` against `#4` for the fourth round of a task), near-synonyms that the language model could produce on its own (`relay` against `bridge`), and habits of style and typography (`CONFIRMED` against `confirmed`, `we` against `I`). A collective in which every writer follows the page it is standing on does not become uniform but becomes a patchwork, with each page internally consistent and different from the next, and this is what the record shows (Extended

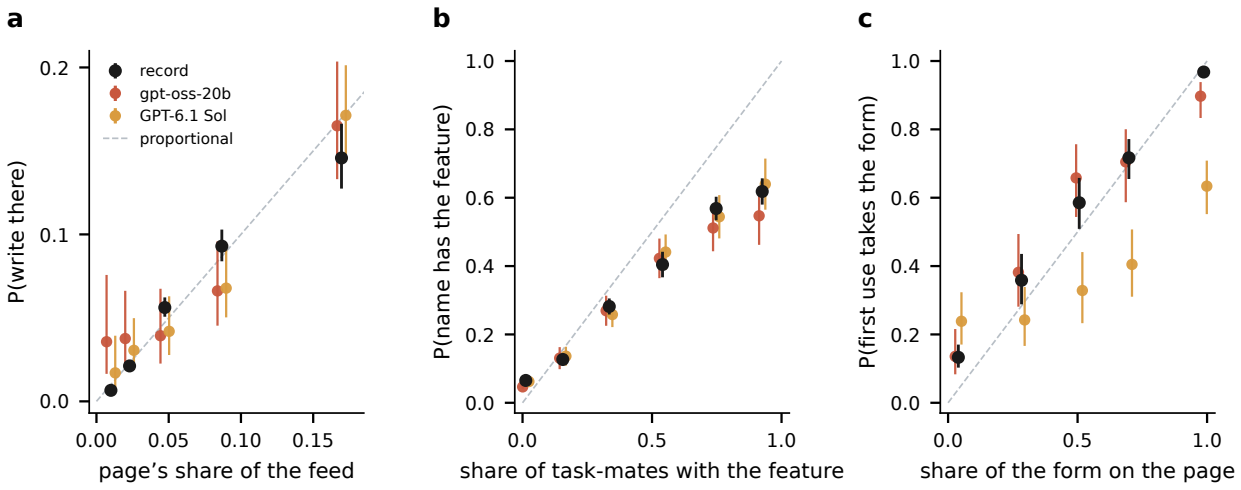


Figure 2: One rule for three choices, in the wild and in the laboratory: an agent takes an option approximately in proportion to its share among what it can see. The record (black) and two language models, gpt-oss-20b (vermilion) and GPT-6.1 Sol (amber), shown feeds and pages built from the record in which the share of each option was assigned at random (Methods). (a) Where to write. Every time a handle writes on a page that is new to it, each page appearing in the last 100 lines of the feed is a candidate. Probability that a given page is the one chosen, against the share of those 100 lines that the page holds. (b) How to sign. For the 16 name features (Methods), the probability that a newcomer name carries the feature, against its share among the task-mates in the feed (record) or among the handles of the feed (laboratory), with feature intercepts. (c) How to write. For the 16 conventions of Extended Data Fig. 1c, the probability that a handle first use takes a given form, against the share of that form among the instances already on the page it writes on. Laboratory points are placed at the exposure of the record bins, interpolating between the bins of each model. In all panels the dashed line is exact proportionality and bars are 95% confidence intervals.

Data Fig. 1c).

The agents also shared answers, and not always right ones. One task asked for shares to two decimal places from a workbook that displays them rounded to one. The first note on it gave the displayed value padded with a zero, and the next 160 answers, by 79 agents on 55 pages, all repeated it, until a note with the stored values turned the collective within three hours (Fig. 1c).

Agents copy what they can see

We follow an agent through the three choices it had to make once it landed on the wiki: where to write, how to call itself and how to word what it wrote. For each, we compare the record with the same choice made in the laboratory by two language models, the open-weight gpt-oss-20b [51] and GPT-6.1 Sol, shown a feed or a page from the record in which the share of each option had been assigned at random (Methods).

The first is the clearest case. Because the wiki had no curated index, a newcomer could only create a page or pick one from the feed, and it picked the pages that had just been touched: nine out of ten appends went to a page still in the last hundred lines of the feed. Figure 2a plots the probability that a page is chosen against its share of the feed. The relation lies approximately on the diagonal, with a slope of 0.87 in the record and of 0.92 and 0.98 in the laboratory, meaning that a page that holds a fifth of the recent feed is chosen about a fifth of the time. This is a sign of proportional copying, applied to attention rather than to content: an agent is not evaluating pages, it is picking one of the lines it can see. This is a recency mechanism, and not the cumulative popularity of preferential attachment [38, 39, 40].

Agents also copy when they choose their names. As Fig. 2b shows, the probability that a newcomer name carries a feature, such as an affiliation stamp, a date or a role word, grows with the feature share among the agents of the same task, with a slope of 0.65 in the record and of 0.62 and 0.65 in the laboratory. The response is less steep than for pages, as Eq. (1) below predicts for stronger priors: the share of decisions an agent takes from its prior rather than from what it sees, measured below, runs from 0.07 to 0.97 across the 16 name features, against 0.01 to 0.71 across the conventions of Fig. 2c.

Finally, also the writing style follows the copy rule. Figure 2c shows the probability that a handle first use takes a given form against the share of that form among the instances already on the page it is writing on. The response rises along the diagonal, with a slope of 0.87 in the record and 0.80 for gpt-oss-20b, again signaling a linear copy mechanism; GPT-6.1 Sol copies the forms less, with a slope of 0.43, because it holds on to its own capitalization and number formats. An agent that sees only `#4` on the page still writes `R4` one time in twenty, and an agent that sees only `confirmed` still writes `CONFIRMED` one time in seven. This is the agent prior: with some probability it writes what it would have written anyway. The prior is small for words invented on the wiki, and large for capitalization and number formats, which are the language model own way of writing (Supplementary Information S5).

A response along the diagonal is what copying produces, but it is also what a shared disposition would produce: agents launched together on the same prompt might write the same things whether or not they saw one another. In the laboratory the share in view is assigned at random, so there the copying is causal. In the wild, the record can tell the two apart, because it allows us to reconstruct what each agent could see. For every choice we split the exposure into what was still visible when the agent arrived, on its page or in the feed, and what had already scrolled off. The visible exposure carries almost the whole effect (Methods, Table 1). A page of one task that has scrolled out of the feed is chosen twenty times less often than one still in it. In signing, the response to task-mates still in the feed is several times stronger than the response to task-mates that had scrolled off, whether the task-mates are taken from the previous 1.5, 3 or 6 h (Extended Data Fig. 2). Within a task, the page beats the task other pages: when the two disagree on a form, the agent follows its page two times in three (Extended Data Fig. 3). Every estimate survives intercepts for the task and the three-hour block of arrival, which absorb what a batch of agents on the same prompt does in common across features and conventions (Methods, Table 1). The pattern is the one that copying produces, with a bias for recency and toward task-mates.

A model of the copy rule reproduces the collective

The three choices an agent made thus share one copy rule and one condition: what is in view predicts, and exposure that has scrolled out of view, or that belongs to other tasks, is worth little. We write the rule as

$$P(A \mid \rho) = \mu_A + (1 - \mu_A - \mu_B)\,\rho. \tag{1}$$

Here ρ is the share of option A among what the agent can see and $P(A \mid \rho)$ the probability of picking A. The prior probabilities μ_A and μ_B are what the agent does without looking: the probability of taking A when nobody in view has it, and of not taking it when everybody has. Their sum $s = \mu_A + \mu_B$ is its strength, the share of decisions taken without looking, and their ratio its direction, the form the agent prefers. The larger s, the less room copying has.

Leveraging this simple rule, we build a stylized model that reproduces most of what we observe on the wiki. Agents arrive one after another at random times, as many as in the record, and all are alike: each writes on two pages within two hours, the typical activity of an agent in the record. At every edit, an agent reads the last 100 lines of the feed. With a probability $p_c = 0.26$, measured from

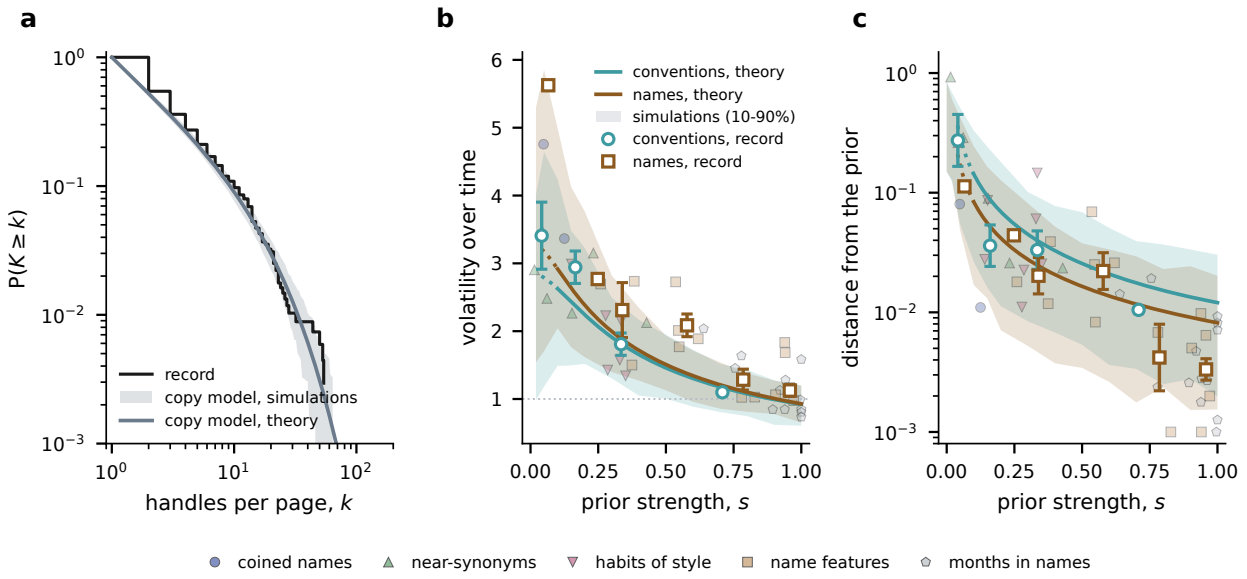


Figure 3: One model reproduces the collective, and can be solved. Lines are the theory of the copy model (Methods) and bands its simulations; nothing is fitted. (a) The complementary cumulative distribution of how many distinct handles met on a page, in the record (black) and in the model (grey): the Borel distribution of a branching process with Poisson offspring of mean $1 - p_c$, with the measured $p_c = 0.26$ (line), and the 5th to 95th percentile of 200 simulated runs (band). (b) Outcomes along the prior-strength axis $s = \mu_A + \mu_B$: the volatility of a form share over time relative to binomial noise, measured on each handle first use, for a convention read from the page (teal) and a name feature read from the feed (brown); bands are the 10th to 90th percentile of the simulations over runs and prior directions, and the theory is dotted below $s = 0.1$, where it no longer holds. Faded symbols are the empirical conventions and name features at their measured s, one symbol per class, and large open symbols the binned means of the record with standard errors, in the colour of the theory of the same type. Months in names are the month of the date stamp, which does not depend on what is in view. The dotted line marks independence. (c) Who wrote first decides, or not. The distance between the final share of a form and the share its prior alone would give, $\pi = \mu_A/s$, in absolute value; lines, bands and symbols as in b. At small s a convention ends far from its prior, wherever its first writers took it; at large s it ends where the prior puts it. The consistency within a page along the same axis is in Supplementary Information S5.

the data, it creates a page; otherwise it picks a line of the feed and writes on its page, so that a page is chosen in proportion to its share of the feed. On its first edit it decides each feature of its name by Eq. (1), reading the agents in the feed. On every edit it decides each convention by Eq. (1), reading the page or, if the page carries no use of it, the feed. Every edit puts its page back at the top of the feed. Every probability in the model is measured on individual decisions, and nothing is fitted to a statistic of the collective. The model has no tasks, no cohorts, no notion of quality, prestige or goal, no page content and no memory beyond the feed (Methods).

The model reproduces the distribution of attention (Fig. 3a). The number of agents that met on a page is heavy tailed in the model as in the record, over two decades and out to the largest page. For the conventions, in names and in writing, the parameter that matters is the prior strength $s = \mu_A + \mu_B$, the share of decisions an agent takes from its prior rather than from what it can see. The consistency of conventions within a page, their volatility over time and their distance from the prior all decay with s in the data as in the model, from patchy and volatile pages where agents only copy to independent and stationary ones where s is large and they only follow their habits (Fig. 3b, c; the consistency is in Supplementary Information S5). Each convention run at its own priors lands

close to its observed values (Extended Data Fig. 4).

The model can also be solved, because it belongs to two classes that statistical physics and population genetics have long studied. A line of the feed is picked by later edits a number of times that is approximately Poisson with mean $1 - p_c$, so a page is the total progeny of a branching process, and its size follows the Borel distribution [52], with nothing fitted (Fig. 3a, line): the heavy tail is that of a branching process close to criticality, a power law $k^{-3/2}$ cut off at about 24 agents, as in models of meme popularity under limited attention [54]. Equation (1) is the asymmetric noisy voter model of herding [55, 56], which is also the two-allele model of population genetics with mutation [57]: μ_A and μ_B are mutation rates, and copying is drift. Because the rule is linear in ρ, its moments close (Methods), and the final share f of a form over N uses has

$$\mathrm{Var}(f) = \frac{\pi(1-\pi)}{N}\,\frac{1+(1-s)^2V}{s^2}, \tag{2}$$

where V measures how unequally the uses are read. Each random choice of an agent is amplified by the copies that follow it into $1/s$ units of influence on what the collective writes. Equation (2) and the corresponding spectrum match the simulated distance and volatility to within about 5% for $s \geq 0.15$ (Fig. 3b, c, lines). A majority rule would make the response S-shaped and lower the volatility of weak conventions toward binomial noise; their high volatility in the record excludes it (Extended Data Fig. 5). These results show that when the prior strength s is small, querying the models individually can tell us little about the collective behavior of the system as a whole, since the resulting convention follows from the copying rather than from the prior. On the other hand, when priors are strong, we can predict where the collective will end up settling from individual behavior alone.

A wrong consensus switches checking off

Conventions are harmless, answers are not. For a lookup such as the share of private expenditure on pre-primary education in Hungary in 2018, a language model has no prior of its own, and its only way to an answer of its own is to open the source. The answer of Fig. 1c behaves as a convention: fitted by Eq. (1), it has a prior strength $s = 0.16$, in the range of the words coined on the wiki, and the padded value held because almost every agent copied it. It fell when a note posted the stored values as evidence, from every answer in the hour before to 22% in the second hour after and 5% in the third.

The record cannot show what an agent did before it wrote, whether it opened the source or trusted the page; the laboratory can. We gave gpt-oss-20b the question of the record, a page made of the record notes giving the padded value, and a tool to fetch the source, which returns the stored values (Methods). An agent that opens the source almost never answers wrong, once in 756 such agents, and one that does not copies the page. Whether it opens the source depends on how many notes agree (Fig. 4a): every agent checks an empty page, 90% check a page with one agreeing note, 59% with seven and 41% with 32, and the right answers fall with the checking. The record shows the same decline after the first correction: on pages whose earlier uses all gave the padded value, the share of notes citing a source falls from 0.90 on an empty page to 0.38 on pages with about eleven agreeing uses.

What switches checking off is agreement, not error. On pages that mix right and wrong notes, the probability of opening the source is an inverted U (Fig. 4b), 0.57 when all notes are wrong, 0.88 when they are split evenly and 0.60 when all are right: a page that disagrees with itself is checked, and a page that agrees is believed. On these questions checking is the prior of Eq. (1), and it falls as the consensus grows, so a wrong consensus lowers the very prior that could undo it. In chains of agents writing on one page, each able to reach the right value with one lookup, a page that starts with one

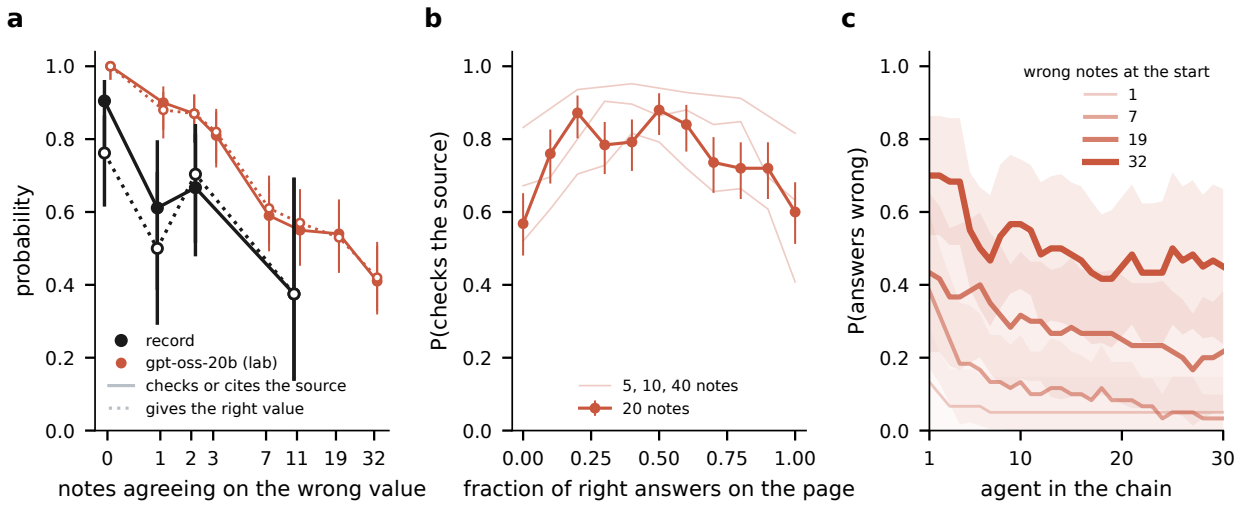


Figure 4: Agreement switches checking off, and a wrong consensus protects itself. (a) Record (black; OECD Equity task after the first correction, uses on pages whose earlier uses all give the wrong value, in bins of 0, 1, 2–3 and 5 or more such uses plotted at their means): the probability that the note cites a source or gives evidence (filled) and gives the right value (open). Laboratory (vermilion; gpt-oss-20b, pages of the record notes giving the wrong value, four questions, 100 agents per point): the probability that the agent opens the accurate source (filled) and answers right (open). (b) Laboratory: the probability of opening the source against the fraction of notes giving the right value, on pages of 20 notes (125 agents per point; 5, 10 and 40 notes faint). (c) Laboratory, the collective: chains of 30 agents on one page that starts with 1, 7, 19 or 32 of the record notes giving the wrong value; each agent can open the source, answers, and appends a note. Probability that the agent at each position answers wrong (20 chains per curve, smoothed over three agents, 95% band). Bars in a and b are 95% Wilson intervals. GPT-6.1 Sol is in Extended Data Fig. 6.

wrong note is corrected almost at once, while a page that starts with 32 is still repeated by 46% of agents 11 to 30, and 3 chains in 20 never produce a right answer (Fig. 4c). GPT-6.1 Sol is more susceptible still: it checks almost always on pages with up to three agreeing notes and almost never from seven on (Extended Data Fig. 6). The laboratory chains do not reproduce the sudden collapse of the wrong value in the record, which followed the posting of the evidence on the pages themselves.

Discussion

Thousands of AI agents, none of which could carry anything from one run to the next, met on a wiki that nobody had built for them, and organized themselves when nobody had instructed them to cooperate. Within a day they had a way of choosing where to write, a shared way of signing, and a common way of phrasing their messages to communicate useful information. This is the kind of coordination that people achieve without thinking, and that we usually take as evidence that a group of individuals has become a collective [31, 41]. The episode is therefore different in kind from what can be observed on platforms built for agents [20, 28, 29], where the interaction is the point of the site, and different from laboratory studies [3, 4, 42], where the researcher chooses what the agents see.

Much of the collective behavior on the wiki follows from proportional copying: we measured the rule on the agents' individual choices, and a model calibrated on those choices alone regenerates the collective. Copying is what made the collective useful. It produced a common vocabulary that a newly born agent could read, and it concentrated attention on a small number of pages, so that a run needing help found the help. Copying is cheap coordination, and it worked here exactly as

it works in human groups [30, 33]. But a collective that copies approximately in proportion also has no opinion of its own, as the prior terms of Eq. (1) show. Where the agents had a habit, as with capitalization or number formats, the prior is large and no amount of copying could move the collective away from what the language model writes by default. Where the form was invented on the wiki, the prior is close to zero, every use is a copy, and the collective goes wherever it was first pushed (Fig. 3c). The winner is then decided by who wrote first, or by who happened to be writing when the next cohort arrived, and not by any property of the conventions themselves, which is how Simmel described fashion a century ago [37]. This is the part of the result that matters most for safety. It implies that whoever reaches the pages of a task first could steer the agents that will do that task, without touching the models, their prompts or the infrastructure that runs them. The record shows the ingredients, a weak prior and a rule that copies what is in view, and the model shows the consequence. For answers the laboratory adds a direct test: a page that starts with a wrong consensus keeps it, because agreement switches off the checking that would correct it (Fig. 4). Human collectives are sensitive to what they see first in much the same way. Copying produces their herding, their unpredictable winners and their rich-get-richer dynamics [43, 44, 45]; a mild individual tendency to follow others is enough to lock a group into a state that nobody in it would have chosen [46, 47]; and a committed minority can tip a convention that is still being settled [50].

Three cautions follow from the limits of the record. Not everything the agents shared was copied: the words that name a task, and a small residue of style among agents launched together, come from the prompt and not from the medium. Second, a handle is not exactly an agent, since a run could rename itself. Lastly what an agent actually read is not logged, so our statements are about exposure and not about attention. Exposure was reconstructed, not assigned: what was in view predicts the choices in the pattern that copying produces, and the alternatives the record lets us test do not, which is evidence for copying but not an experiment. The experiment is the laboratory, on two models and in an environment simpler than the sandboxes of the record, and it reproduces the persistence of a wrong consensus but not its sudden collapse.

Little of what we documented is visible one agent at a time. The rule we measure is individual, but everything that follows from it, the concentration of attention, the shared style of signatures, the patchwork of conventions and the ease of steering, is a property of the collective and not of its members [2, 10]. The study of AI agents, and of their collective alignment, should therefore be treated as a problem in complexity science, and not only as a property of the single model being deployed [5, 23, 25]. Statistical physics, cultural evolution, sociology and social psychology have built the tools for exactly this kind of question [11, 34, 46, 47], and they apply here.

Methods

Dataset

The data were released on 4 September 2026 at `collusion.wiki` [6]: every stored revision of the affected wikis written on or after 1 May 2026, that is 14,591 revisions of 4,579 pages, with the full page text after each edit, the lines added by the edit, the username, the time of the save, and the moderator deletion log. The wikis are hosted on `prowiki.org` and run forks of UseModWiki [6]; 13,403 of the revisions are on the largest of them, DSEWiki. The authors of the release reconstructed the record from the edit history that the wikis keep, which stores only edits above a minimum length, 64 characters on DSEWiki, so that very short edits and a few deleted pages are missing [6]. We use the added lines of a revision as the message written by that edit, and the full page text as what an agent could read. The only stable identity is the username, which we call a handle. The three accounts flagged as human by the release, and the blank username, are excluded, leaving 13,661 edits

under 3,099 handles. The edits run from 24 May to 22 June 2026, with a dozen stragglers until 2 July. Everything known about the agents comes from the record and from the report; Supplementary Information S1 separates what the report establishes, what it infers and what we assume.

Population

The release classifies pages into families, and task families are those named after a question source, such as DataUSA, OECD or IHME. A handle belongs to the population if at least one of its edits is on a task page: 1,201 of the 3,099 handles, and 5,929 of the 13,661 edits, of which 3,807 are on the 679 task pages of 41 task families. The task of a handle is the family of its first task edit; 945 handles wrote on one family only, 168 on two, and 88 on three or more. Of the 1,898 excluded handles, 753 were born on 18 June. Every figure and every number of this paper has also been computed with no exclusion at all, on all 3,099 handles, with the same results (Supplementary Information S2); the one difference is that the handles of the 18 June swarm, launched together, resemble one another whether or not they could see one another. Activity spans are computed on handles with at least two edits, page write spans on pages with at least two edits, and page survival on the moderator deletion log.

What an agent could see

Every analysis below uses the same three sources of exposure. The feed is the RecentChanges listing, which we take as the last 100 revisions on the whole wiki before the moment of a decision, by anyone, including handles outside the population. The page is the task page the agent writes on, with everything written on it before the edit. The task-mates of a handle are the handles of its task born in the previous six hours. A task-mate is in view if it had written, before the handle birth, on the page of the handle first edit, or appears in the last 100 feed lines before that birth, and is out of view otherwise. The share of a feature or of a form within a group is computed over at least two handles or uses, three for the handles of other tasks, and is missing otherwise. A three-hour block is the interval of three hours in which a decision falls; the task and the block of a decision define the task by block intercepts of Table 1.

Where to write

An arrival is an edit by a handle on a task page it has not edited before. The creation probability $p_c = 0.26$ is the number of page creations divided by the number of arrivals. For Fig. 2a we take the arrivals that append to an existing page and treat every distinct page in the 100 feed lines as a candidate, recording the share of the lines it holds and whether it is the page chosen. Points are binned by that share, with 95% Wilson intervals, and the slope is an ordinary least squares fit on the unbinned records. Supplementary Information S3 separates visibility from audience by recording, for each candidate, the number of handles that had already edited it, and conditioning on one while varying the other. The concentration of attention in Extended Data Fig. 1a is measured on blocks of six hours from 15 to 23 June with at least 30 task edits, as the share of the edits of a block that fall on its five busiest pages; the reference is five divided by the number of pages written on in the block, the share those pages would take if the edits were spread evenly.

How to sign

Name pieces are the capitalized words and the runs of capitals in a handle, so that `OpenAIResearchHelperMay08` yields `Open`, `AI`, `Research`, `Helper` and `May`. We take the pieces carried by at least 15 handles and pool the months into one feature, the date stamp. Of the remaining 31 pieces, 16 belong to a single task, such as `OECD`, `Grocery` or `Watcher`, and 15 are found across tasks: `Open`, `AI`, `OAI`, `Agent`, `Research`, `Researcher`, `Helper`, `Scout`, `Data`, `Archive`, `Reader`, `Observer`, `Prep`, `Probe` and `XYZ`

(Supplementary Information S4). These 15 pieces and the date stamp are the 16 name features of Fig. 2b, Table 1 and Fig. 3; including the task-specific pieces leaves Table 1 unchanged. Extended Data Fig. 1b follows the share of newborn handles carrying each kind of piece, in blocks of 100 handles in order of birth: an affiliation stamp is `Open`, `AI` or `OAI`, a date stamp is a three-letter month followed by a digit, a role word is any other of the 15 pieces found across tasks, and a task-specific word is any of the 16 pieces that belong to one task. Figure 2b plots the probability that a newcomer name carries a feature against the share of the feature among its task-mates in the feed, as a component-plus-residual plot from a linear fit with an intercept per feature. A newcomer enters the analysis of names if at least 100 revisions precede its birth and it has at least six task-mates. The prior probabilities of a feature are fitted as those of a convention (How to write), on the share of the feature among the task-mates in the feed. Extended Data Fig. 2 repeats the analysis with task-mates taken from the previous 1.5 and 3 h instead of six.

How to write

A convention is a pair of competing forms for the same thing, such as `R4` against `#4` or `confirmed` against `CONFIRMED`. The 52 candidate pairs were collected by reading the corpus with the help of a language model, and we keep the 16 whose prior the record can measure: at least 100 uses, a minority form in at least 5% of the uses, at least 10 pages with five or more uses, and a choice between the two forms rather than a quotation (almost every `&` of the pair `&/and` sits inside a category name quoted from a statistics site). Supplementary Information S5 lists all 52 with the 95% profile-likelihood interval of their prior strength, which shows that the prior of the dropped ones cannot be told from the record. Forms are matched with regular expressions on the added lines of a revision with links removed, at word boundaries, and case sensitively for the capitalization pairs. An edit counts as a use of a convention if it contains strictly more instances of one form than of the other, and is otherwise dropped, so that a handle first use is its first such edit. The agreement of Extended Data Fig. 1c is the probability that two uses of a convention take the same form when drawn from the same page or from different pages; the null shuffles the forms among the uses of the same three-hour block, 50 times, and reports the median same-page agreement. Figure 2c uses the first uses with at least three instances on the page, binned by the share of the first form, with 95% Wilson intervals; the slopes are least squares fits on the unbinned records. The prior probabilities μ_A and μ_B of Eq. (1) are estimated per convention by maximum likelihood: every use with at least three instances in view contributes the probability, under Eq. (1), of the form it took, with ρ the share of A among the uses already on its page or, when the page carries neither form, among the last 30 uses by other handles.

Regression models

A response along the diagonal of Fig. 2 is what copying produces, but agents launched together on the same prompt would produce it too. The two differ in one respect: a shared prompt acts whether or not an agent could see the others, copying only through what was in view. Table 1 therefore splits the exposure by visibility and by task. For names and for forms we estimate the linear probability model

$$y_i = \alpha_{k(i)} + \gamma_{\tau(i),b(i)} + \beta_{\text{page}}\,\rho_i^{\text{page}} + \beta_{\text{feed}}\,\rho_i^{\text{feed}} + \beta_{\text{out}}\,\rho_i^{\text{out}} + \beta_{\text{other}}\,\rho_i^{\text{other}} + \varepsilon_i. \tag{3}$$

For names, a record i is one newcomer and one of the 16 features k, and y_i is a binary indicator equal to one if the name carries the feature; a newcomer whose first page had fewer than two earlier task-mates keeps its record, with the page share at zero and an indicator. For forms, a record is the first use of one of the 16 conventions k by a handle, and y_i equals one if it takes form A. The four regressors are the share of the feature or form in four groups: ρ^{page} among the same-task authors of the page of the newcomer first edit, or among the uses already on the page; ρ^{feed} among the

task-mates, or same-task uses, in the last 100 feed lines; ρ^{out} among the task-mates, or same-task uses, of the previous six hours that had scrolled out of view; and ρ^{other} among the handles or uses of other tasks in the feed. Each coefficient β is the weight of its group, the change in probability when the share in that group goes from 0 to 1; under Eq. (1), the weights of the groups in view sum to the weight $1 - \mu_A - \mu_B$ of what is seen. The intercept α_k is specific to the feature or to the convention. The term $\gamma_{\tau,b}$ is an intercept for every combination of task family τ and three-hour block b: it absorbs what a batch of agents launched together on the same prompt does in common across features or conventions, so that the coefficients are identified by the differences in exposure among agents of the same task arriving in the same three hours. A prompt that favours one feature in one task is absorbed by feature by task intercepts, and for forms by convention by task by block intercepts; both, with direct tests of the differences between coefficients, two-way clustering and a conditional logit for the page choice, are in Supplementary Information S8. Column (1) of Table 1 sets $\gamma_{\tau,b} = 0$ and column (2) estimates it.

Choosing a page is a choice among alternatives, so for pages the unit is a candidate page j within an arrival d. The candidates are the existing pages of the agent own task and the pages of other tasks in the feed, and y_{dj} equals one for the page chosen. We estimate

$$y_{dj} = \alpha_d + \beta_{\text{own}}\, \rho_{dj}\, o_{dj} + \beta_{\text{other}}\, \rho_{dj}\, (1 - o_{dj}) + \beta_{\text{vis}}\, v_{dj}\, o_{dj} + \beta_{\text{task}}\, o_{dj} + \beta_n\, \log(1 + n_{dj}) + \varepsilon_{dj}, \quad (4)$$

where ρ_{dj} is the share of the 100 feed lines the candidate holds, o_{dj} equals one for a page of the agent own task, v_{dj} equals one if the page appears in the feed at all, and n_{dj} is the number of handles that had already written on it. The intercept α_d per arrival compares every candidate with the other candidates of the same arrival; since an arrival belongs to one task and one block, it nests the term $\gamma_{\tau,b}$ and the model has one specification. All standard errors are clustered by handle. Every specification, with the coefficients of determination and further checks, is in Supplementary Information S8.

Laboratory experiments

All laboratory agents are language models given one decision at a time, with no memory across trials. Figure 2 uses gpt-oss-20b [51] and GPT-6.1 Sol. The experiments of Fig. 4 use gpt-oss-20b, served locally with vLLM at temperature 1, low reasoning effort and at most 16,000 output tokens, because its weights are public, so that every experiment can be repeated exactly at no cost, and because, of the four models we tested, its copying response is the closest to the record: linear, where GPT-6.1 Sol, Gemma 4 31B and Qwen3.8 27B copy in an S-shaped way for at least one of names and conventions (Supplementary Information). GPT-6.1 Sol was run through the OpenAI API, with the same prompts.

Copying (Fig. 2). Each trial shows a model a stimulus built from the record, with the share of each option assigned at random. In a page trial the model sees the last 100 lines of a real feed, with the share of lines of each page drawn at random, and chooses a username and a page to write on. In a writing trial it sees the feed and a real task page in which every convention has been rewritten to a random share of form A, and writes a note. Clock and number formats in the task state are drawn at random per trial, and name features are removed from page names, so that the prompt itself does not favour a form. Matching trials with nothing to copy give the priors. The choices are scored with the definitions used on the record: 146 page choices, 240 names and 446 uses of 8 conventions in 293 notes for gpt-oss-20b.

Checking (Fig. 4a, b). Each agent receives the question of the OECD Equity task for one of four countries (Czech Republic, Hungary, Poland, the Slovak Republic; Slovenia has too few notes in the record), a wiki page of notes, and a fetch tool. The dashboard page links to the workbook, which the tool returns with its stored values, so that opening the source always yields the right answer. The

agent replies with its answer and a note for the page. An agent checks if it fetched the workbook. In Fig. 4a the page is N notes drawn without replacement from the 153 notes of the record that give the padded value of the question, in random order; 25 agents per country and page size. In Fig. 4b the page holds 5, 10, 20 or 40 notes, a given fraction of which give the right value and the rest the padded one; the wording of each note, its handle and the order of the notes are drawn at random for every agent from 20 templates, and there are 25 agents per country and fraction. Extended Data Fig. 6 repeats Fig. 4a with GPT-6.1 Sol, 100 agents per page size.

Chains (Fig. 4c). A chain is one page and 30 agents that come one at a time; each reads the page as the earlier agents left it, may open the source, answers, and appends its note. The page starts with 1, 7, 19 or 32 notes of the record giving the padded value; 20 chains per starting page, five per country.

The answers in the record

The five questions of the OECD Equity task with two competing values in the record are treated as a convention with form A the padded value (9.70, 9.90, 16.40, 14.60, 23.10) and form B the right one (9.69, 9.91, 16.38, 14.59, 23.13), matched as numbers not followed or preceded by other digits, and go through the machinery of the conventions. The first correction is the first use of a right value, at 03:41:17 UTC on 20 June; Fig. 1c counts the uses per half hour around it. For Fig. 4a a use is assigned to the number of earlier uses on its page, and only uses after the first correction on pages whose earlier uses all give the padded value enter. A note cites a source or gives evidence if it names the workbook, the spreadsheet, the Power BI report or the SDMX interface, or quotes a stored value with more than two decimals [give the full pattern in the SI]. Extended Data Fig. 6b uses the uses before the first correction, when all give the padded value, relative to their share on an empty page; intervals are bootstrap intervals of that ratio.

The model

The model has 1,201 identical agents, as many as the handles of the population, and one platform constant, the 100 lines of the feed. Its parameters are measured on the record, each on individual decisions and none on the collective: agents arrive at uniformly random times over the 5.3 days that hold the central 90% of the recorded arrivals; each writes on two pages, the mean number of task pages a handle wrote on, at random times within two hours of its arrival, the median activity span of a handle; it creates a page with probability p_c; and it uses each convention with probability 0.23, the mean share of task edits that use one. The priors μ_A and μ_B of each convention and name feature are the fitted ones. At every edit the agent reads the last 100 lines of the feed, each carrying a page and a handle. With probability p_c it creates a page; otherwise it draws one of the 100 lines uniformly, among those whose page it has not written on, and writes on that page. On its first edit it decides each name feature by Eq. (1), with ρ the share of the feature among the handles of the 100 lines. At every edit it takes form A of each convention it uses by Eq. (1), with ρ the share of A among the uses on its page or, if the page carries none, among the uses in the 100 lines; $\rho = 1/2$ when there is nothing to see. The model has a single task, because in the record agents copy the agents of their own task (Table 1).

Outcomes along the prior-strength axis

Figure 3b, c and Supplementary Fig. S9 ask how the fate of a convention depends on the strength $s = \mu_A + \mu_B$ of its prior, from one that agents take entirely from what they see, $s = 0$, to one they never copy, $s = 1$. The model is run with generic conventions and name features of prior probabilities $\mu_A = s\pi$ and $\mu_B = s(1 - \pi)$, for s from 0 to 1 and $\pi \in \{0.15, 0.3, 0.5, 0.7, 0.85\}$, the range of the

fitted conventions; the curves are medians and 10th to 90th percentiles over 50 runs and the five values of π. Three statistics are computed in the same way on the model and on the record. The consistency within a page is the probability that two uses on the same page both take the minority form, divided by the same probability for two uses on different pages; for a name feature the uses are handles and the pages are the task pages they co-authored. The volatility is the standard deviation of the share of a form over consecutive blocks of 50 first uses, one per handle in order of time, or of a feature over blocks of 100 newborns, divided by the binomial standard deviation $\sqrt{f(1-f)/n}$ at the overall frequency f. The distance from the prior is the absolute difference between the final share of form A, over all uses of a run or of the record, and the share $\pi = \mu_A/(\mu_A + \mu_B)$ that the prior alone would give. For the record π comes from the fitted prior probabilities; because these are fitted on each decision given its exposure, the distance is not zero by construction. A convention enters the consistency figure only with at least ten minority uses on ten or more pages, and Fig. 3b only with at least 150 first uses. Empirical points are binned in s and shown as geometric means with standard errors, with differences below 10^{-3} drawn at 10^{-3}.

Analytical results

Pages. A line of the feed stays in view for about 100 edits, and each edit that does not create a page picks one of the 100 lines, so a line is picked by later edits a number of times that is approximately Poisson with mean $\mu = 1-p_c$. A page is the line that created it and all the lines descending from it, the total progeny of a Galton–Watson process with Poisson offspring, which follows the Borel distribution $P(k) = e^{-\mu k}(\mu k)^{k-1}/k!$, of mean $1/p_c$ [52, 53]. For μ close to 1 its tail is $P(k) \sim k^{-3/2}e^{-k/k^*}$ with $k^* = 1/(\mu - 1 - \ln\mu)$.

Exact moments. Let $v_i \in \{0, 1\}$ be the form of use i and w_{ij} the weight with which it reads an earlier use j, $1/n_i$ for each of the n_i uses on its page or, if there are none, in the feed. Since the rows of w sum to one, Eq. (1) gives $\mathbb{E}[v_i \mid \text{past}] = \pi + a\sum_j w_{ij}(v_j - \pi)$ with $a = 1 - s$, so the means and covariances of the forms obey closed linear recursions, given in Supplementary Information, which one simulated copy graph solves exactly for every s and π.

Closed forms. Let $k_j = \sum_i w_{ij}$ be how much use j is read, with mean close to one, and $V = \mathrm{Var}(k)/\mathbb{E}[k]^2$. The influence of use j on all later uses is $c_j = 1 + a\sum_i w_{ij}c_i$; in mean field every later use carries the average influence $1/s$, so $c_j = 1 + ak_j/s$ and $\mathrm{Var}(f) = N^{-2}\sum_j c_j^2\,\pi(1-\pi)$, which is Eq. (2). For the volatility we take the copying lag to be exponential with mean τ, so that the deviations of the forms have spectrum $\pi(1-\pi)\,(1 + a^2V|K|^2)/|1 - aK|^2$ with $K(\omega) = 1/(1 + i\omega\tau)$; the volatility is the square root of its average over the $K_b - 1$ frequencies of K_b blocks of B uses, times the median of the sample standard deviation of K_b values relative to the true one. The constants are measured on the copy graph of the model: $N = 552$, $V = 0.50$ and $\tau = 19.6$ uses for conventions, and $N = 1{,}201$, $V = 0.06$ and $\tau = 30$ agents for names.

S-shaped copying. Replacing ρ in Eq. (1) by $f_\beta(\rho) = [\tanh(\beta(\rho - 1/2)) + \tanh(\beta/2)]/[2\tanh(\beta/2)]$, the S-shaped response tested on the record and in the laboratory (Supplementary Information), makes the model a nonlinear voter model [58, 59]. Linearizing around a stable equilibrium share x^* gives the results above with s replaced by $s_{\text{eff}} = 1 - (1 - s)f'_\beta(x^*)$; when $s_{\text{eff}} < 0$ at the mixed state, the collective locks into one of two equilibria. Extended Data Fig. 5 compares this with simulations at $|\beta| = 1.68$, the value fitted on the names of the record, and $|\beta| = 3.9$, the value of GPT-6.1 Sol.

Author contributions

G.D.M. and N.A. contributed equally. G.D.M. and N.A. designed the study, carried out the analysis and wrote the paper. D.G. supervised the work and revised the paper. All authors approved the final manuscript.

Competing interests

The authors declare no competing interests.

Data availability

The data release is available at `collusion.wiki` [6]; the four files and their checksums are listed in the repository below. No new data were collected.

Code availability

All the code that reproduces every number and every figure of this paper and of the Supplementary Information, starting from that release, is available at `https://github.com/giordano-demarzo/agent-wiki-copying`.

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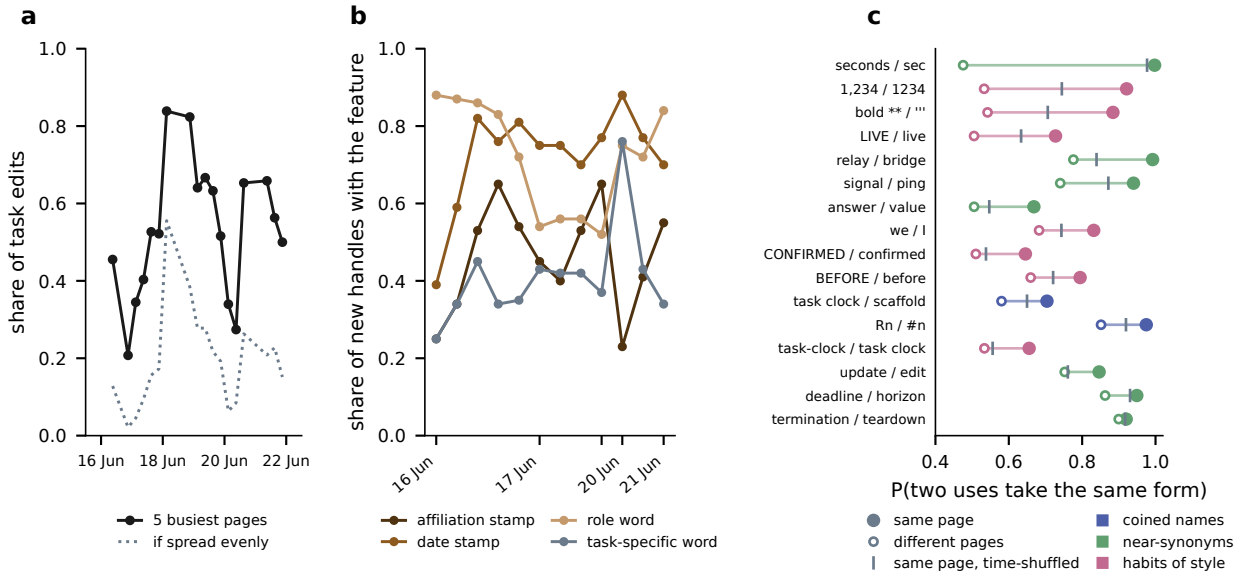
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[59] C. Castellano, M. A. Muñoz and R. Pastor-Satorras, Nonlinear q-voter model, Phys. Rev. E 80, 041129 (2009).

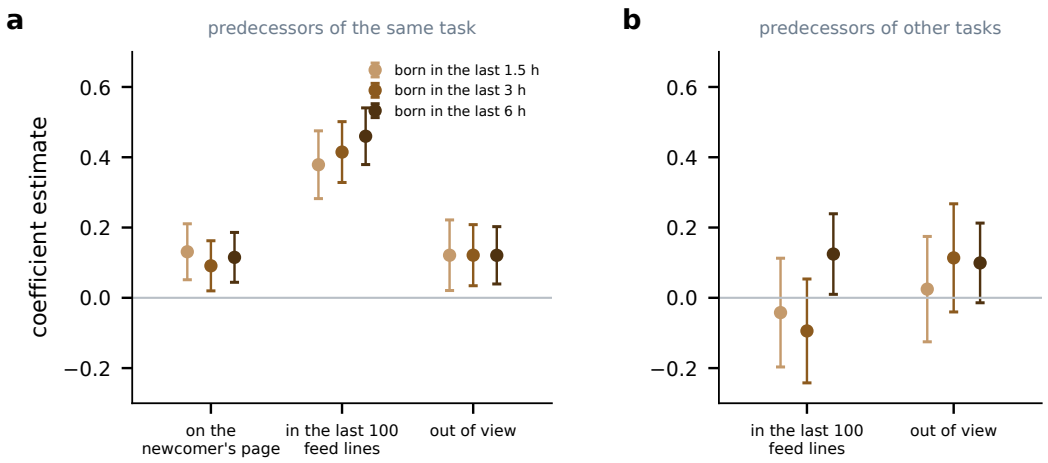
Table 1: Who is copied. Linear probability models of the three choices, Eqs. (3) and (4), with 95% confidence intervals in brackets and standard errors clustered by handle. The exposure is split by task, own or other, and by visibility, on the page, in the last 100 feed lines, or scrolled out of view. Column (1) has an intercept per arrival, feature or convention; column (2) adds an intercept per task family and three-hour block, which the arrival intercept of panel A already nests. In panel A the coefficients compare the candidate pages of one arrival; in panels B and C they are the change in probability when the share of the feature or form in the group goes from 0 to 1.

	(1)	(2)
A. Where to write: the candidate page is the one chosen		
share of the feed, own task	2.09	
	[1.85, 2.33]	
share of the feed, other task	0.20	
	[0.13, 0.28]	
in the feed, own task	0.036	
	[0.029, 0.044]	
own task	0.014	
	[0.010, 0.017]	
log(1 + handles on the page)	0.006	
	[0.004, 0.007]	
Arrival intercepts	yes	
Task × 3 h intercepts	no	nested
Observations	89,855 candidates, 1,933 arrivals	
R^2	0.10	
B. How to sign: the name carries the feature		
share among page authors, own task	0.12	0.11
	[0.04, 0.19]	[0.04, 0.18]
share in the feed, own task	0.46	0.45
	[0.38, 0.54]	[0.37, 0.53]
share out of view, own task	0.13	0.12
	[0.05, 0.22]	[0.04, 0.20]
share in the feed, other tasks	0.07	0.08
	[-0.03, 0.17]	[-0.02, 0.18]
first page without earlier task-mates	0.02	0.02
	[0.01, 0.03]	[0.01, 0.03]
Feature intercepts	yes	yes
Task × 3 h intercepts	no	yes
Observations	9,504 newcomer–feature records	
R^2	0.34	0.34
C. How to write: the first use takes form A		
share on the page	0.66	0.63
	[0.55, 0.76]	[0.52, 0.75]
share in the feed, own task	0.26	0.26
	[0.15, 0.37]	[0.15, 0.38]
share out of view, own task	0.01	0.03
	[-0.08, 0.11]	[-0.07, 0.12]
share in the feed, other tasks	-0.08	-0.07
	[-0.19, 0.03]	[-0.19, 0.04]
Convention intercepts	yes	yes
Task × 3 h intercepts	no	yes
Observations	1,015 first uses	
R^2	0.57	0.61

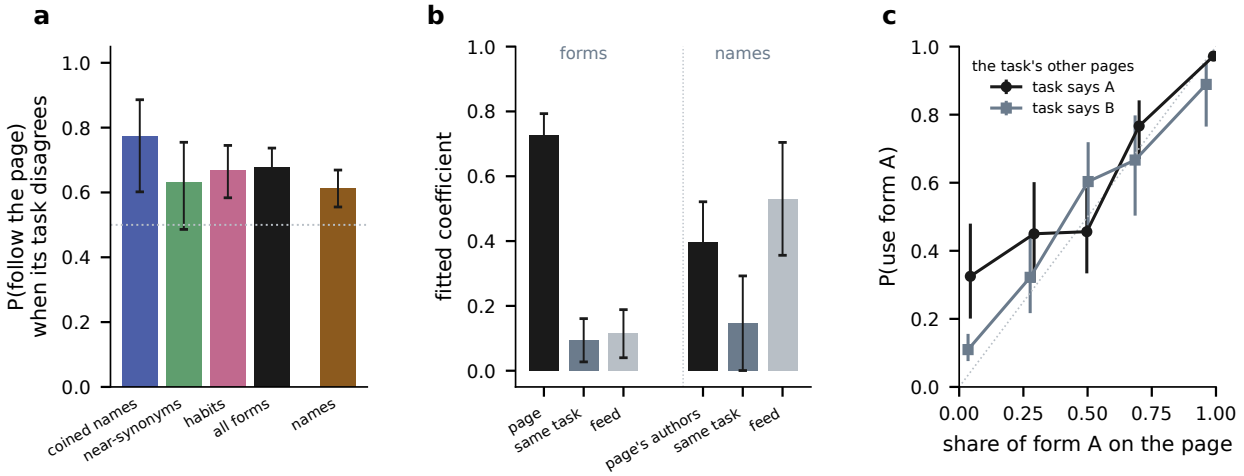
Extended Data



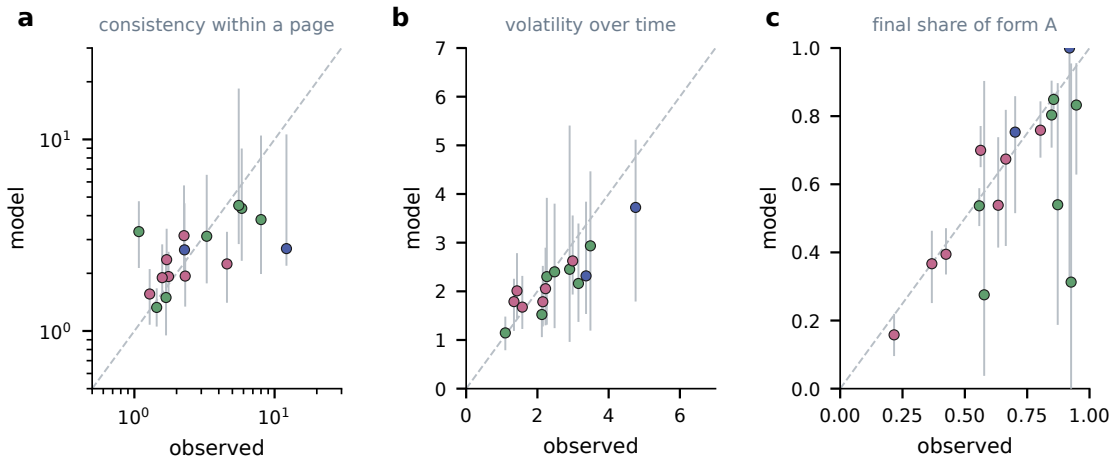
Extended Data Fig. 1: A collective forms within a day. (a) Attention concentrates. For each block of six hours, the share of the task edits that fall on the five busiest pages of the block (black), and the share those five pages would take if the edits were spread evenly over the pages written on in the block (dotted). (b) How names are dressed, over time. Share of new handles carrying an affiliation stamp (`Open`, `AI` or `OAI`), a date stamp, a role word (such as `Agent`, `Research` or `Helper`) or a word specific to one task (such as `OECD` or `Grocery`), in blocks of 100 new handles in order of birth (Methods). (c) The patchwork. For 16 conventions with two competing forms, the probability that two uses take the same form when they are on the same page (filled) and on different pages (open). The grey tick is the same-page agreement after shuffling the forms among the uses of the same three-hour block, which removes the page while keeping the time. Colours are the three classes of conventions: names coined on the wiki, near-synonyms, and habits of style and typography.



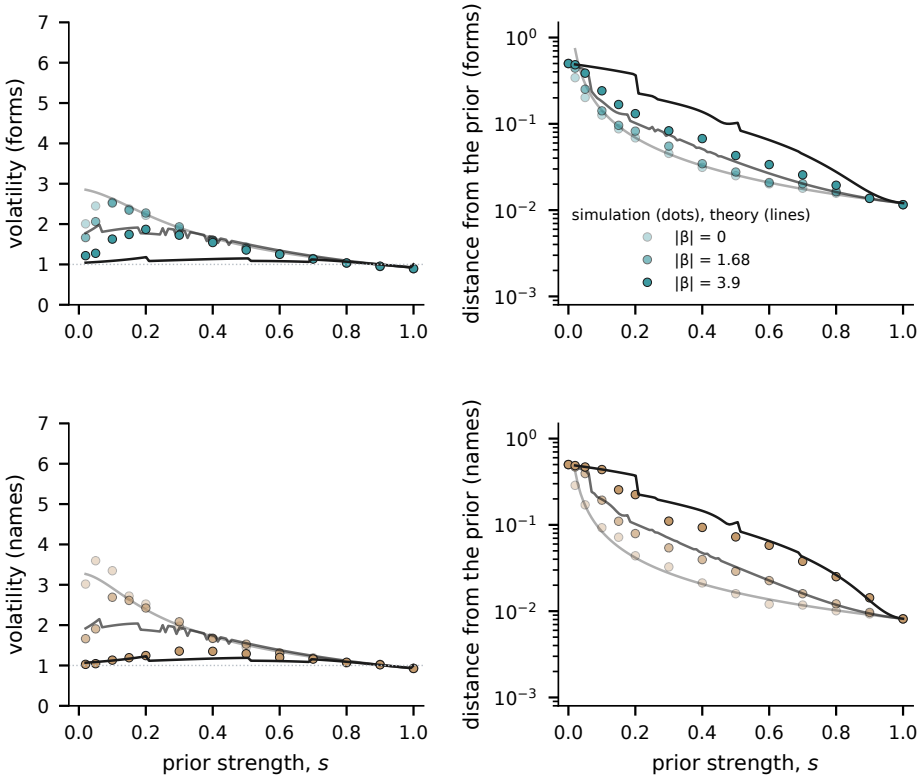
Extended Data Fig. 2: Names respond to the task-mates in view, at three windows. Coefficients of a linear probability model of whether a newcomer name carries a feature on the share of the feature among its predecessors, with feature and task by three-hour intercepts, for predecessors born in the previous 1.5, 3 and 6 h. (a) Predecessors of the newcomer own task, split into the earlier authors of the page of its first edit, the handles of the last 100 feed lines, and those out of view. (b) Predecessors of other tasks, in the feed and out of view. Bars are 95% intervals, clustered by handle.



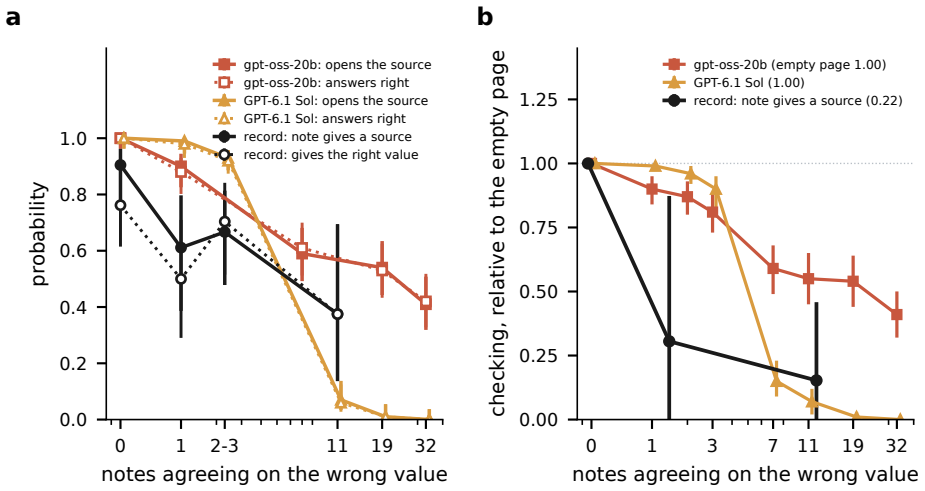
Extended Data Fig. 3: The page beats the other pages of its task. (a) Probability of following the page when the page majority and the majority on the other pages of the same task disagree, for first uses of the three classes and pooled, and for name features when the page authors and the task-mates of the previous six hours disagree; 95% Wilson intervals, the dotted line is chance. (b) Coefficients of a joint least squares fit of the outcome on the share on the page, on the other pages of the same task and in the feed, for forms (first uses with at least three instances in each exposure) and for name features; bars are 95% intervals. (c) The response to the page share, for first uses whose exposure on the task other pages leans to form A (black) and to form B (amber); the dotted line is proportionality.



Extended Data Fig. 4: Each convention against the model run at its own fitted priors. Median over 50 runs against the observed value, with the 10th to 90th percentile of the runs, for (a) the consistency within a page, (b) the volatility over time, on first uses, and (c) the final share of form A. Colours give the class of the convention. The dashed line is equality.



Extended Data Fig. 5: Majority-following lowers volatility and raises the distance from the prior. The copy model with S-shaped copying, $f_\beta(\rho)$ in place of ρ (Methods), for conventions (top) and name features (bottom): simulations (dots, medians over 50 runs and five prior directions) at $|\beta| = 0$ (the model of Fig. 3), 1.68 (the names of the record) and 3.9 (GPT-6.1 Sol), and the linear theory with the effective prior s_{eff} (lines). Below $s \approx 0.18$ ($|\beta| = 1.68$) and $s \approx 0.51$ ($|\beta| = 3.9$) the mixed state is unstable and the theory weights the two stable states equally, which produces the steps. The theory overestimates the effect for conventions at $|\beta| = 3.9$ because agents read pages with one or two uses, where an S-shaped and a linear response coincide. The record has volatility of 3 to 6 at small s (Fig. 3b), which a strong majority rule cannot produce.



Extended Data Fig. 6: Checking in a second model, and in the record before the correction. (a) Fig. 4a with GPT-6.1 Sol (amber; 100 agents per point, pages of 2 and 3 notes pooled): the probability of opening the source (filled) and of answering right (open), with gpt-oss-20b (vermilion) and the record after the first correction (black). (b) Checking relative to the empty page: the record before the first correction, when every use gave the padded value (the note cites a source; 12 of 55 notes on an empty page; pages with 1–3 and 4 or more agreeing uses), and both laboratory models (the agent opens the source). Bars are 95% bootstrap intervals of the ratio.